Euclid's Elements — Book I
Proposition 21

CGJteamLab

If from the ends of one of the sides of a triangle two straight lines are constructed meeting within the triangle, the straight lines so constructed are together less than the remaining two sides of the triangle, but contain a greater angle.

1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle.

Let ABC be a nondegenerate triangle. Let D be an interior point, and let the line BD meet the side AC at E, so that

$$A - E - C \qquad \text{and} \qquad B - D - E.$$

Euclid asserts simultaneously that

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

The proposition is therefore the first result in this part of Book I whose reconstruction naturally separates into two substantial comparison arguments: one for segments and one for angles.

The formal reconstruction preserves this separation. The segment inequality is obtained by combining Proposition I.20 with a synthetic "addition of unequal segments" argument. The angle inequality is proved independently, using Proposition I.16 and the strict angle comparison machinery developed in the preceding propositions.

A particularly important feature of the formalization is that no numerical length function is introduced. Expressions such as $BA+AC$ and $BD+DC$ are represented by ordinary constructed segments.

2 The Classical Configuration

Let ABC be a nondegenerate triangle. Choose a point D inside the triangle and let the line BD meet AC at E.

Thus

$$A - E - C, \qquad B - D - E.$$

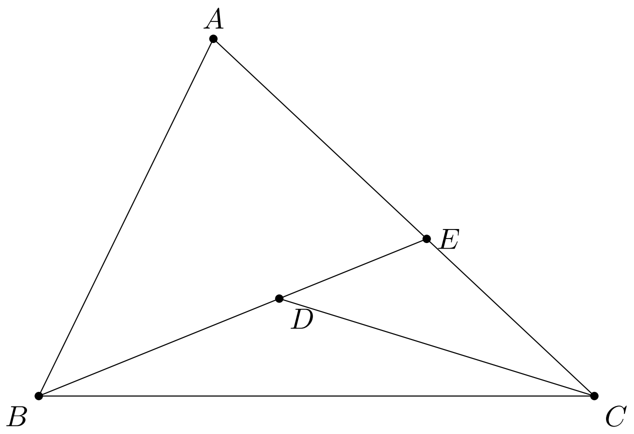

Figure 1: Classical configuration for Proposition I.21. The point D lies inside triangle ABC; the line BD meets AC at E.

For the segment inequality, Euclid first compares the broken line $BA + AE$ with BE by applying Proposition I.20 to triangle BAE. The remaining pieces EC and the corresponding copied segment are then added to obtain the comparison with $BA + AC$.

For the angle inequality, the point E allows the exterior-angle theorem to be applied to appropriate auxiliary triangles. The resulting strict angle comparisons are transported along the collinear rays determined by A, E, C and B, D, E.

The formal proof makes these two uses of the same configuration explicit.

Wilkins's presentation is especially useful here because the same diagram supports two logically independent chains. For segments, Proposition I.20 gives

$$DC < DE + EC, \qquad DE < DA + AE,$$

and therefore

$$BD + DC < BD + DE + EC = BE + EC < BA + AE + EC = BA + AC.$$

For angles, the same point E exposes a clean two-step exterior-angle chain:

$$\angle BDC > \angle BEC > \angle BAC.$$

The single classical diagram above therefore serves both parts of the proof. There is no mathematical gain in repeating the same triangle a second time. I.21 is instead a useful example of one configuration supporting two distinct arguments: a segment-sum comparison and an angle-order comparison.

3 Classical Proof

Since D lies inside triangle ABC, the line BD meets AC at an interior point E.

Apply Proposition I.20 to triangle ABE. It gives

$$BE < BA + AE.$$

Add EC to the two sides in the geometric sense. Since

$$AE + EC = AC,$$

we obtain

$$BE + EC < BA + AC.$$

Now apply Proposition I.20 to triangle CDE. Since $B - D - E$,

$$BD + DE = BE.$$

Proposition I.20 gives

$$DC + DE > CE,$$

or equivalently

$$CE < DC + DE.$$

Combining the appropriate segment comparisons in the classical configuration yields

$$BD + DC < BA + AC.$$

For the angle part, the exterior-angle theorem I.16 is applied to the triangles determined by the interior point and the two transversals. It yields strict comparisons which, after identifying the collinear rays, give

$$\angle BAC < \angle BDC.$$

Thus both assertions of I.21 follow:

$$BD + DC < BA + AC \qquad \text{and} \qquad \angle BAC < \angle BDC.$$

The classical proof is extremely compact because segment addition, subtraction, and replacement of collinear rays are read directly from the diagram. These are precisely the operations that must be made explicit in the formal reconstruction.

4 From Euclid's Diagram to the Formal Argument

The classical proof compresses several different kinds of reasoning into a single diagram. The formal reconstruction separates them.

First, statements such as "the whole is greater than the part" are represented by the strict synthetic relation `HilbertSegmentLess`. Second, sums and differences of segment lengths are not numerical expressions: they are realized by actual collinear configurations and transported by congruence. Third, the angle comparison needed in the proof is obtained from the already formalized exterior-angle machinery rather than inferred from the appearance of the figure.

Thus the reconstruction preserves Euclid's geometric strategy while making explicit the segment arithmetic and order-theoretic steps that the classical presentation leaves implicit.

5 Synthetic Segment Arithmetic

The segment part of I.21 exposes an important feature of the Hilbert reconstruction.

The formal development does not introduce a numerical length function, nor does it define an algebraic addition operation on segments. Instead, a sum is represented by a constructed segment.

For example, a point X satisfying

$$B - A - X \qquad \text{and} \qquad AX \cong AC$$

makes BX a representative of

$$BA + AC.$$

Similarly, a point Y satisfying

$$B - D - Y \qquad \text{and} \qquad DY \cong DC$$

makes BY a representative of

$$BD + DC.$$

Consequently the desired inequality is represented formally by

$$BY < BX.$$

The formal representatives can be pictured independently of the classical triangle:

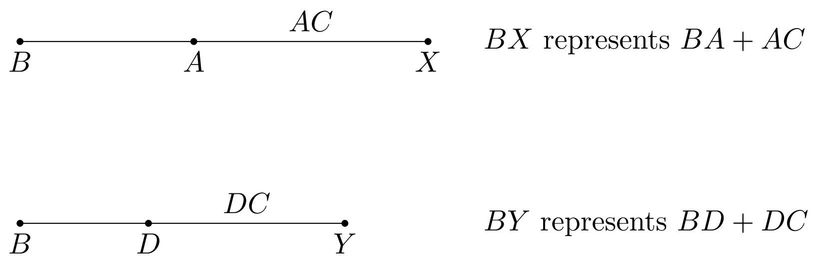

Figure 2: Synthetic representatives of the two segment sums in the Lean statement. The target inequality is the ordinary strict segment comparison $BY < BX$.

In Lean this is

```
HilbertSegmentLess Geo B Y B X
```

This observation made an explicit predicate for segment sums unnecessary. The existing language of betweenness, congruence, and strict segment comparison is sufficient.

This is not merely a simplification of I.21. It is evidence that a useful fragment of segment arithmetic can be expressed internally in the synthetic language of Book Zero.

6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three major components.

```
                  common configuration
               A-E-C,  B-D-E
                       |
           +-----------+-----------+
           |                       |
           v                       v
     segment branch           angle branch
           |                       |
        I.20 twice              I.16 twice
           |                       |
 construct representatives    BAC < BEC < BDC
    of BA+AC, BD+DC                |
           |                       |
           v                       v
       BY < BX             angle BAC < BDC
           \                       /
            \                     /
             +---------+---------+
                       |
                       v
               euclid_proposition_21
```

The diagram is deliberately architectural rather than metric. In particular, the points X and Y in the formal theorem are not additional features of Euclid's printed diagram; they are witnesses that reify the two segment sums inside the synthetic language.

6.1 The Segment Helper

The first reusable component is

`euclid_proposition_21_segment_helper`

It begins with

$$A - E - C, \qquad B - A - H,$$

together with

$$AH \cong AE$$

and

$$BE < BH.$$

A point F is constructed on the ray AH with

$$AF \cong AC.$$

Since $AE < AC$, congruence transport gives

$$AH < AF.$$

The Book Zero order machinery then yields

$$A - H - F.$$

Together with $B - A - H$, this produces

$$B - A - F \qquad \text{and} \qquad B - H - F.$$

Thus BF represents $BA + AC$.

Next a point G is constructed beyond E with

$$EG \cong EC.$$

Hence

$$B - E - G,$$

and BG represents $BE + EC$.

From

$$A - E - C, \qquad A - H - F,$$

and

$$AE \cong AH, \qquad AC \cong AF,$$

the theorem

```
bookZero_differenceOfParts
```

gives

$$EC \cong HF.$$

The inequality $BE < BH$ can therefore be extended by congruent terminal parts. The theorem

```
bookZero_53_lessThanAdditive
```

yields

$$BG < BF.$$

This is the synthetic form of

$$BE + EC < BA + AC.$$

6.2 The Angle Branch

The angle comparison is isolated as

```
euclid_proposition_21_angle
```

with conclusion

```
HilbertAngleLess Geo B A C B D C
```

representing

$$\angle BAC < \angle BDC.$$

The proof first reconstructs the noncollinearity information required for the auxiliary triangles. This is a substantial formal task: Euclid simply reads these facts from the diagram.

Proposition I.16 then supplies the strict exterior-angle comparisons. The results are transported through collinear and same-ray configurations, and strict angle comparison is propagated to the required pair of angles.

The essential point is that no numerical angle measure is used. The proof remains entirely inside the synthetic relation `HilbertAngleLess`.

6.3 Final Segment Comparison

The main theorem first applies Proposition I.20 to triangle BAE. It obtains a point H with

$$B - A - H, \qquad AH \cong AE,$$

and

$$BE < BH.$$

The segment helper extends this inequality to a comparison of constructed representatives.

The remaining part of the proof constructs a point Y satisfying

$$B - D - Y, \qquad DY \cong DC.$$

The point X on the other side satisfies

$$B - A - X, \qquad AX \cong AC.$$

After a sequence of congruence transports and applications of the Book Zero comparison lemmas, the proof reaches

```
hBY_BX :
    HilbertSegmentLess Geo B Y B X
```

which is exactly

$$BD + DC < BA + AC.$$

7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_21
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hAEC : Geo.Between A E C)
    (hBDE : Geo.Between B D E) :
    (exists X Y : Geo.Point,
      Geo.Between B A X /\
      Geo.Congruent A X A C /\
      Geo.Between B D Y /\
      Geo.Congruent D Y D C /\
      HilbertSegmentLess Geo B Y B X) /\
    HilbertAngleLess Geo B A C B D C := by
```

The existential part is the exact synthetic encoding of the segment inequality.

The conditions

```
Geo.Between B A X
Geo.Congruent A X A C
```

say that BX represents $BA + AC$.

Likewise

```
Geo.Between B D Y
Geo.Congruent D Y D C
```

say that BY represents $BD + DC$.

The comparison

```
HilbertSegmentLess Geo B Y B X
```

therefore expresses the first assertion of I.21.

The second assertion appears directly as

```
HilbertAngleLess Geo B A C B D C
```

The two branches are deliberately proved separately and assembled only at the end.

8 Relationship with Earlier Propositions

8.1 Proposition I.16

Proposition I.16 provides the exterior-angle inequalities needed for the angle branch.

This is the principal earlier proposition behind the proof of

$$\angle BAC < \angle BDC.$$

The formal reconstruction also reuses the same strict-angle language that was introduced and strengthened in the development surrounding I.16–I.18.

8.2 Proposition I.20

Proposition I.20 is fundamental to the segment branch.

Applied to triangle BAE, it supplies the initial inequality

$$BE < BA + AE.$$

In the formal proof the sum $BA + AE$ is represented by a constructed point H:

$$B - A - H, \qquad AH \cong AE,$$

so that

$$BE < BH.$$

This inequality is then enlarged synthetically by adding congruent parts.

8.3 The Cumulative Character of I.21

Proposition I.21 is markedly more cumulative than the immediately preceding propositions.

Its proof is no longer dominated by one construction. Instead it requires a coordinated use of order, congruence, segment comparison, ray transport, angle comparison, I.16, I.20, and numerous Book Zero lemmas.

For this reason I.21 is a useful test of whether the preceding formal development has produced a genuine working language rather than a collection of isolated theorem statements.

9 Hilbert and Book Zero Components Used

9.1 HilbertSegmentLess

Strict segment comparison supplies a synthetic replacement for numerical inequalities between lengths.

Its witness-based definition also makes it possible to recover interior points and to turn inequalities into betweenness configurations.

9.2 Congruence transport

The proof repeatedly changes representatives of the same segment. This requires strict inequalities to be transported through segment congruence.

Among the relevant Book Zero results are

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```

These operations play the role that substitution of equal numerical lengths would play in an algebraic proof.

9.3 Difference of parts

The theorem

```
bookZero_differenceOfParts
```

expresses the synthetic subtraction principle.

From two collinear decompositions with congruent wholes and congruent initial parts, it identifies the remaining parts.

In I.21 it produces

$$EC \cong HF.$$

9.4 Addition of inequalities

The theorem

`bookZero_53_lessThanAdditive`

is especially significant here.

It permits an existing strict segment inequality to be extended by congruent parts. In I.21 it turns

$$BE < BH$$

into the corresponding comparison after EC and HF have been added.

This is one of the clearest examples so far of Book Zero functioning as a synthetic arithmetic of segments.

9.5 Segment-Inequality Transitivity

The theorem

`bookZero_52_lessThanTransitive`

allows the several intermediate comparisons to be composed into the final inequality.

9.6 Sum of parts

The existing theorem

`bookZero_sumOfParts`

is used when two adjacent congruent pieces must be recognized as giving congruent whole segments.

Together with difference-of-parts and inequality transport, this provides much of the elementary segment calculus required by I.21.

10 What I.21 Reveals About Book Zero

The reconstruction of I.21 gives a useful answer to a structural question about the project.

It might appear natural to introduce explicit objects representing formal sums of segments. Indeed, an intermediate version of the development contained such a definition.

The completed proof shows that this is not presently necessary.

The combination of

$$\text{betweenness} + \text{congruence} + \text{strict segment comparison}$$

already permits sums to be constructed, compared, enlarged, reduced, and transported.

Thus the expressions

$$AB + CD, \qquad AB < CD,$$

need not force the development prematurely into an algebra of lengths.

The geometric witnesses themselves carry the arithmetic.

This is important for the intended role of Book Zero. Its purpose is not merely to shorten proofs. It supplies the intermediate operations by which ordinary synthetic reasoning can be expressed without abandoning the primitive Hilbert geometry.

I.21 is the strongest example of this role encountered so far.

11 Dependency Summary

The segment branch can be summarized schematically as

$$\begin{array}{c}
A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear} \\
\downarrow \\
\text{I.20 on } BAE \\
\downarrow \\
BE < BA + AE \\
\downarrow \\
\text{construct congruent remainders} \\
\downarrow \\
\texttt{bookZero_differenceOfParts} \\
\downarrow \\
\texttt{bookZero_53_lessThanAdditive} \\
\downarrow \\
\text{congruence transport and transitivity} \\
\downarrow \\
BD + DC < BA + AC.
\end{array}$$

The angle branch is

$$\begin{array}{c}
A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear} \\
\downarrow \\
\text{noncollinearity and ray reconstruction} \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\text{strict angle transport} \\
\downarrow \\
\angle BAC < \angle BDC.
\end{array}$$

Finally,

$$\begin{tikzcd}
BD + DC < BA + AC \arrow[dr] & & \angle BAC < \angle BDC \arrow[dl] \\
& \texttt{euclid_proposition_21}. &
\end{tikzcd}$$

No new geometric axiom is introduced.

12 Formalization Notes

Several points that are nearly invisible in the classical proof become substantial in Lean.

First, every assertion that a point lies on the same line, on the same ray, or on the opposite continuation of a ray must be derived explicitly.

Second, the notation $AB + AC$ cannot silently denote a number. A concrete point must be constructed whose position and congruence properties make an ordinary segment represent that sum.

Third, inequalities cannot simply be rewritten by replacing equal lengths. Congruence transport requires explicit theorems.

Fourth, the two assertions of I.21 genuinely have different proof architectures. Keeping them separate in the formal development makes their dependencies visible and produces reusable intermediate results.

The resulting Lean proof is much longer than Euclid's proof, but the additional length is not caused by a different mathematical idea. It records the incidence, order, congruence, and transport operations which the classical diagram suppresses.

13 Conclusion

Proposition I.21 is an important transition point in the formalization of Book I.

The classical proposition combines a comparison of broken lines with a comparison of angles. Its proof assumes a mature ability to manipulate segment sums, strict inequalities, collinear configurations, and exterior angles.

The Lean reconstruction preserves this synthetic structure. It does not replace Euclid's argument by coordinates, numerical lengths, or angle measures.

More importantly, the reconstruction demonstrates that the Book Zero layer is beginning to perform its intended function. Results such as difference of parts, addition of inequalities, congruence transport, ray conversion, and strict-comparison transitivity combine into a usable language in which a substantially more complicated Euclidean argument can be expressed.

The fact that no separate `HilbertSegmentSum` object is needed is part of this conclusion. Synthetic segment arithmetic can, at least at this stage, remain geometry rather than become an external algebra of lengths.

The Hilbert reconstruction of Euclid I.21 is complete. It introduces no new geometric axiom, and the complete development through Proposition I.21 compiles successfully.